LIBERO-PeRM: Benchmarking Personalized Robotic Manipulation

Li et al. · NeurIPS 2026 Datasets & Benchmarks · arxiv.org/abs/2610.22372

TL;DR

Finishing a task and finishing it **the way you want** are two different problems. This paper builds a benchmark that measures them separately: 5 task suites, 312 tasks and 40.8K episodes. Today's VLA models complete the tasks but mostly ignore user preferences. Fine-tuning teaches the preferences, but generalizing to new scenes and combining several preferences are still hard.

Background you need

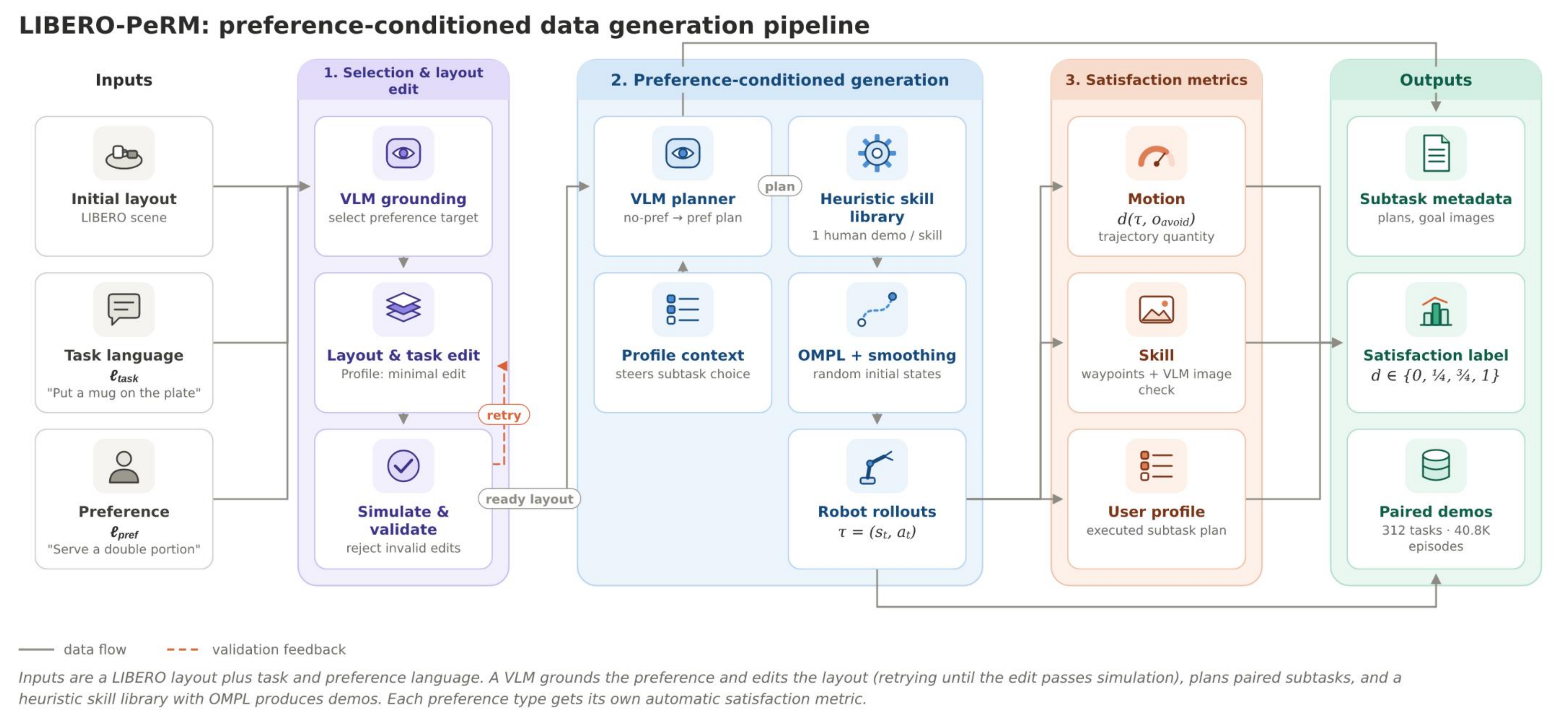
- **VLA (Vision-Language-Action) model**: takes camera images and a text instruction, and outputs robot actions directly.
- **LIBERO**: a widely used simulation benchmark for robot manipulation. This paper builds on top of it.
- **SR (success rate)**: share of episodes where the task is done. **SA (preference satisfaction)**: how well the preference was followed, counted only when the task succeeds.
- **LoRA / SFT**: lightweight fine-tuning that trains a small part of a big model.

The problem

The same command, "put the mug on the table", can mean handle facing out for one person and on a coaster for another. Existing benchmarks only score **task success**, so there was no way to measure whether a robot follows personal preferences. Prior work also defined and evaluated "personalization" in incompatible ways.

What they did

- **A preference taxonomy with 3 types**: Motion (e.g. slow down, avoid an object), Skill (e.g. pick by the handle) and User profile (e.g. prefers dairy). That gives 37 single preferences plus 18 combinations.
- **An automatic data pipeline**: a VLM minimally edits the scene, writes paired plans with and without the preference, and a heuristic skill library plus the OMPL planner generates the demos.
- **An automatic scorer per preference type**: trajectory math (Motion), waypoints plus a VLM image check (Skill), and the executed subtasks (User profile).



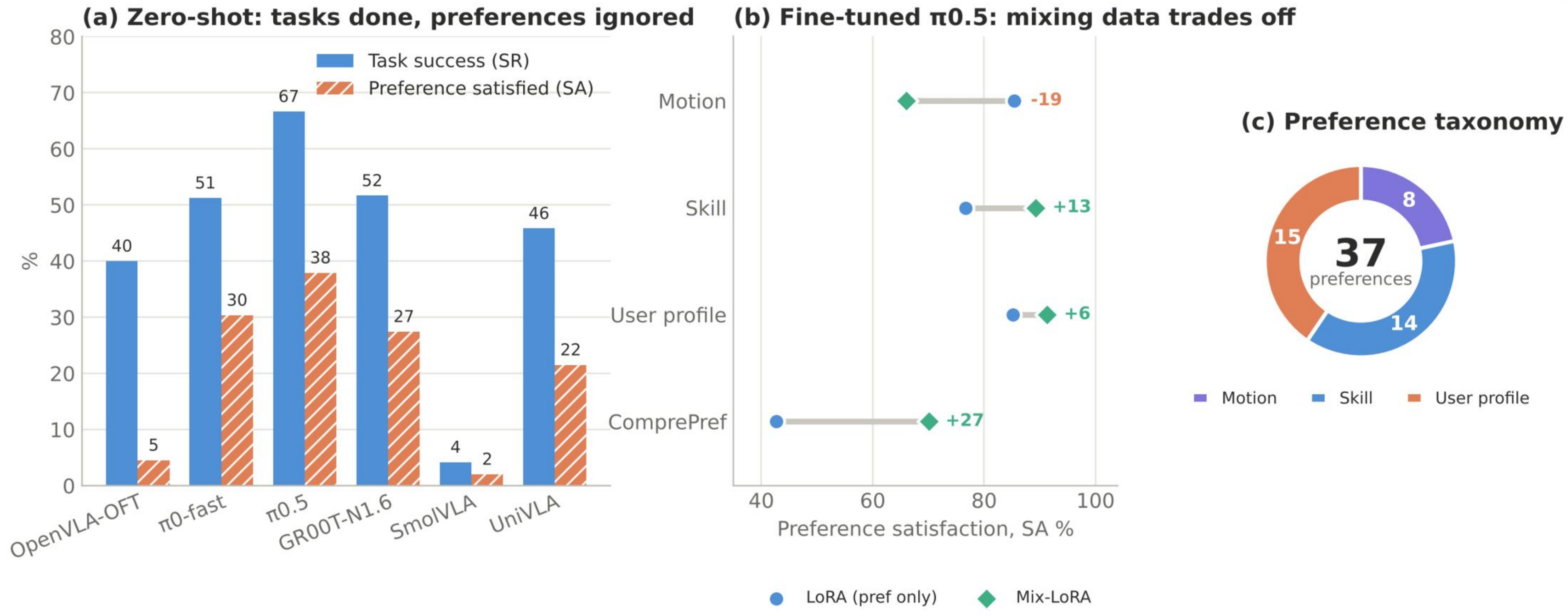
LIBERO-PeRM: preference-conditioned data generation pipeline

Inputs are a LIBERO layout plus task and preference language. A VLM grounds the preference and edits the layout (retrying until the edit passes simulation), plans paired subtasks, and a heuristic skill library with OMPL produces demos. Each preference type gets its own automatic satisfaction metric.

Figure 1. Data generation pipeline. Inputs → (1) preference selection and layout edit, retried until it passes simulation → (2) preference-conditioned generation → (3) per-type satisfaction metrics → outputs.

Key findings

- **Zero-shot fails**: π0.5 completed Motion and Skill tasks 97.50% and 100.00% of the time, but satisfied the preference only 24.38% and 25.00% of the time. It does the job and ignores the preference.
- **Fine-tuning makes preferences learnable**: LoRA on the preference data raises All-Pref SA to 82.48%.
- **Mixed training helps, at a cost**: Mix-LoRA keeps 91.00% no-preference SR, versus 52.33% for sequential training, but Motion SR drops from 90.00% to 66.88%.
- **Diversity beats volume**: at the same number of demos, more distinct layouts helped more than more demos per layout.
- **Combining preferences is still hard**: composing two preferences reaches 25.0%, and pulling one preference back out of a combination gets 0.0% SA.



(a) Zero-shot: tasks done, preferences ignored

(b) Fine-tuned π0.5: mixing data trades off

(c) Preference taxonomy

Figure 2. (a) Zero-shot: preference satisfaction (SA) is far below task success (SR). (b) Fine-tuned π0.5: SA for LoRA vs Mix-LoRA. (c) The 37 preferences by type.

Limitations

- Simulation only, with no real-robot experiments.
- Only preferences that can be stated in words and measured automatically. Inferring preferences from interaction history is out of scope.
- Combinations stop at 2 preferences, and the benchmark doesn't test detecting conflicts or asking the user.
- Human validation is a 15-person survey.

Why it matters

Home robots need to go beyond "did the task" to "did it **my way**". This benchmark separates task success from preference satisfaction and gives VLA research a common yardstick for comparing and improving personalization.